

2. Peter, Claire and David compared the hardness of three samples of water, **A**, **B** and **C**. They added soap solution to each sample until it gave a lather on shaking. They carried out a fair test.

(a) Give **two** ways they made it a fair test. [2]

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(b) Their results are shown in the table.

Water sample	Volume of soap solution needed to give a lather (cm ³)			
	Peter	Claire	David	Mean
A	18	19	17	
B	8	10	9	9
C	1	1	1	1

(i) Calculate the mean volume of soap solution needed for sample **A** to give a lather. **Show your working.** [2]

Mean = cm³

(ii) Which of the three water samples is the **hardest**? Give a reason for your answer. [2]

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(c) The box shows some advantages and some disadvantages of hard water.

strengthens bones and teeth	forms limescale when heated
wastes soap	reduces heart disease
forms a scum with soap	tastes better

Choose **all** the disadvantages of hard water from the box. [2]

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(d) Choose from the box an ion that causes hardness in water. [1]

K ⁺	Na ⁺	Ca ²⁺	Cu ²⁺
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